

Tutorial 5: WRES1201 – Computer System Architecture

1. What are the key properties of semiconductor memory?
2. Explain why one type of RAM is considered to be analog and the other digital.
3. What are some applications for ROM (in scope of internal memory)?
4. What are the differences among EPROM, EEPROM and flash memory?
5. List the main differences between SRAM and DRAM.
6. Why do DRAMs generally have much larger capacities than SRAMs constructed in the same fabrication technology?
7. Describe the task of 'S' in SDRAM.
8. What are the two errors category in semiconductor memory system?
9. Suppose an 8 bit data word stored in memory is 11000010. Using the Hamming algorithm, determine what check bits would be stored in memory with the data word. Show how you got your answer.

1.
 - It is made using semiconductor material
 - It is usually used as internal memory, such as RAM, ROM and cache
 - It supports random access, meaning any memory location can be accessed directly.
 - It may be volatile or non-volatile
 - It has fast access speed compared to external storage.

2. - DRAM is considered analog because it stores data as electrical charge in a capacitor. The stored charge gradually leaks away, so DRAM must be refreshed gradually.

- SRAM is considered digital because it stores data using flip-flop circuits. A flip-flop has two stable states, representing 0 and 1. Therefore, SRAM does not need refreshing as long as power is supplied.

3.
 - Firmware
 - BIOS
 - Microcontroller programs.
 - Lookup tables
 - System startup instructions.

4.

Type	Erasing Method	Main features
EPROM	Ultraviolet light	must usually be removed from the system to erase
EEPROM	Electrical signals	can erase and rewrite individual bytes
Flash memory	Electrical signals	erase data in blocks, faster and denser than EEPROM

5.

Feature	SRAM	DRAM
Storage element	Flip-flop	Capacitor
Refresh needed	No	Yes
Speed	Faster	Slower
Cost	More expensive	Cheaper
Density	Lower	Higher
Capacity	Smaller	Larger
Typical use	Cache memory	Main memory

6. This is because DRAM has simpler memory cell structure which is usually one transistor and one capacitor while SRAM cell usually uses several transistors, commonly 6 transistors. Since DRAM cell is smaller, more cells can fit into the same chip area. Therefore, DRAM can achieve much larger capacity than SRAM using the same fabrication technology.

7. The "S" in SDRAM means Synchronous. Its task is to synchronize DRAM operation with system clock. This allows memory transfers to be coordinated with the processor timing, improving performance compared with asynchronous DRAM. SDRAM also supports burst mode, where a series of data bits can be transferred rapidly after the first access.

- 8.
- Hard errors : permanent physical faults in the memory hardware.
 - Soft errors : temporary errors caused by events such as electrical noise or radiation.

9. 11000010

$$2^r \geq m+r+1$$

$$m=8$$

$$r=1 \quad 2^1 \geq 8+1+1$$

$$2 \geq 10$$

$$r=2 \quad 2^2 \geq 8+2+1$$

$$4 \geq 11$$

$$r=3 \quad 2^3 \geq 8+3+1$$

$$8 \geq 12$$

$$r=4 \quad 2^4 \geq 8+4+1$$

$$16 \geq 13$$

(4 check bits needed)

$$11000010 = D_1 D_2 D_3 D_4 D_5 D_6 D_7 D_8$$

Even parity

Position	1	2	3	4	5	6	7	8	9	10	11	12
Bit type	C1	C2	D1	C4	D2	D3	D4	C8	D5	D6	D7	D8
Value	?	?	1	?	1	0	0	?	0	0	1	0

C₁

C₂

C₄

C₈

$$1+1+0+0+1 = 3 \text{ (odd)} \quad 1+0+0+0+1 = 2 \text{ (even)} \quad 1+0+0+0 = 1 \text{ (odd)} \quad 0+0+1+0 = 1 \text{ (odd)}$$

$$C_1 = 1$$

$$C_2 = 0$$

$$C_4 = 1$$

$$C_8 = 1$$

Check bits = 1011

Stored Hamming Code = 101110010010